# Emotion Detection in Social Media with Retrieval-Augmented Generation (RAG) Justification

Bachelor Semester Project 5

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ABSTRACT

The project was developed by student **João Bernardo Sousa Faria**, under the supervision of their tutor **Salima Lamsiyah**. The project aims to build a system that detects emotions (e.g., anger, joy, sadness) in social media posts (Reddit comments) and explains its predictions. Instead of being a black box, the system will use Retrieval-Augmented Generation (RAG) to justify the detected emotion by retrieving and presenting similar posts from emotion-labeled datasets. For example, a post classified as *anger* will be accompanied by evidence such as: *"Similar posts expressing anger also use terms like 'furious' and 'disgusted'."*

## Introduction

Emotion detection in social media has become an important task in natural language processing, with applications ranging from mental health monitoring to opinion analysis. Social media texts are usually short, informal, and emotionally charged in different ways, which makes automatic emotion classification challenging. While modern deep learning models can achieve strong predictive performance, their decisions are often **difficult to understand**, limiting trust and transparency in real-world applications.

Most emotion detection systems focus mainly on improving classification accuracy, which offers limited insight into the reason that explains the prediction of a specific emotion. The lack of explainability locally present is problematic in domains where users and scientists, for example, may need understandable justifications for model outputs. As a result, there is a growing necessity for approaches that detect emotions accurately **and** provide human-interpretable explanations for their predictions.

In this work, an **emotion detection system for social media** is proposed. It combines supervised classification with retrieval-augmented generation (RAG) to produce evidence-based justifications. To achieve the goal, fine-tuning a RoBERTa-based multi-label classifier on the GoEmotions dataset is required to predict multiple emotions from an input post. To improve prediction reliability, a **per-label decision threshold optimization** is applied on the validation set. For justification, semantically similar social media posts are retrieved from the training corpus and used as external evidence to explain the predicted emotions. The justification is generated by linking emotional cues in the input post to emotions expressed in the retrieved examples, resulting in an example-based, evidence-grounded explanation.

The main contributions of this paper are threefold:

- A robust multi-label emotion classification model for social media text;
- An evaluation-driven threshold tuning strategy that significantly improves classification performance;
- A retrieval-augmented justification framework that provides transparent, example-based explanations for emotion predictions.

The remainder of this paper is structured as follows:

- Related Work: Citation of existing works from the literature which are relevant to the project;
- Scientific Background: Definition of all scientific concepts used to implement the project;

- Methodology: Description of all the steps of the chosen approach;
- Experimental Results: Description of the dataset and evaluation measures, summary of the obtained results in tables, discussion and analysis of the mentioned findings;
- Conclusion and Future Work: Conclusion of the project, as well as discussing its limitations, and suggesting directions for future work.

# Related Work

In this project, the primary piece of existing work used for project development advancements is the paper about the dataset used for the project, namely "GoEmotions: A Dataset of Fine-Grained Emotions". The paper focuses on the purpose of the dataset, as well as presenting its specifications. The paper addresses the abundance of datasets focused on sentiment, respectively positive and negative, and are single-label. With this, the GoEmotions dataset offers a variety of emotions (more specifically **28 fine-grained emotion categories**, neutral label included) and around **58,000 Reddit comments** with multiple combinations of these, resulting in the dataset being identified as a multi-label dataset. Furthermore, the BERT-based model used for the development of the paper achieved an average F1-score of **0.46** across their proposed taxonomy, leaving room for noticeable improvement. (1)

# Scientific Background

## Emotion Detection in Text

Emotion detection in text allows for the identification of emotional states in the written environment. Unlike sentiment analysis, where there are two possible outcomes (positive and negative), this concept focuses on recognizing specific emotions such as happiness, sadness, or anger.

Especially in social media, emotion detection is more challenging than usual due to informal language, abbreviations, short texts and implicit emotional cues. Furthermore, most posts express multiple emotions, independently of the amount of text, which strongly encourages the urge for multi-label approaches. (2)

## Multi-Label Classification

As mentioned previously, the project requires a multi-label approach to achieve its designed goal. Multi-label classification is the reason why an input can get assigned many labels simultaneously.

1. Usually, models output a probability for each label (which can be explained by their respective frequency in the dataset) with the use of a **sigmoid activation function**.
2. With this, decision thresholds are then necessary to identify the predicted emotions.

Since emotion datasets are often imbalanced, per-label thresholds are preferred to massively improve classification performance. (3)

## Transformer-Based Language Models

Transformer-based language models are neural structures that contain self-attention mechanisms that capture contextual relationships between words. In other words, they create contextualized text representations which fit well for understanding subtle language.

To improve its performance on the social media text, RoBERTa (the chosen pre-trained transformer) is fine-tuned on the GoEmotions dataset, resulting in the model adapting its representations for emotion detection. (4)

## Semantic Text Representations & Similarity

Semantic text representations encode the meaning of a text into a vector located in a predefined dimension. This means that texts with similar meanings are somewhat close to each other in the embedding space. The similarity between texts is often measured with **cosine similarity**.

Comparison of many texts is also enabled by semantic text representations, which also sets the basis for semantic retrieval. (5)

## Information Retrieval & Evidence-Based Explanation

Identifying texts which are considered relevant to a given query is the role of information retrieval methods. In the context of this project, retrieval is applied on social media posts that are semantically similar to the input post. These can also be used as external evidence, since these usually have at least one emotion in common, as well as similar language patterns which are associated with specific emotions. In this project, this procedure aligns with an example-based explanation strategy. (6)

## Retrieval-Augmented Generation (RAG)

RAG is the product of combining retrieval mechanisms with natural language generation.

1. The retriever selects relevant text structures (ex: documents, examples, etc.) from an external corpus (in the context of the project, it is the GoEmotions dataset);
2. The generator then formulates an output conditioned on the input as well as the information obtained from retrieval.

It is important to note that the purpose of RAG in the project is to generate justifications for the emotions which were predicted thanks to explanations in the retrieved examples. RAG was not required for improving prediction accuracy, since its only objective is justifying the obtained output. (7)

## Explainability via Example-Based Justification

Explainability ensures that model predictions are understandable to humans. Therefore, instead of explaining model parameters, an example-based justification is opted.

As hinted in previous sections, predictions are justified through posts which express the same emotions as the input post or similar, which accompanies human reasoning by analogy and returns instinctive, evidence-grounded explanations for the revealed emotion predictions. (8)

# Methodology

## Dataset and Preprocessing

**Dataset:** The experiments were done in the presence of the **GoEmotions dataset**, consisting of:

- **27 fine-grained emotion categories** and an **additional neutral label**, as well as approximately **58,000 Reddit comments**.
- Each comment can be associated with one or more emotions, classifying the task as a multi-label classification problem.

**Preprocessing:** The dataset is divided into 3 splits following the original partitioning:

- Training
- Validation
- Test

Text preprocessing is **minimal** to preserve the informal and expressive characteristics of social media language.

- Comments are tokenized individually thanks to the pretrained transformer tokenizer used in the project.
- Emotion annotations get converted into multi-hot binary vectors which correspond to the label set. In the end, every vector contains 28 values (with each value corresponding to one emotion label), which are unique for every post.

## Emotion Classification Model

- The model responsible for performing emotion classification is a **transformer-based language model**. It is optimal for multi-label classification, due to its capability of attaching a task-specific classification head which outputs one score for each emotion label.

  If an emotion is detected in a comment, then the value below the respective emotion label is 1. Otherwise, if a label is not detected in the comment, then its value is 0.

  The sigmoid activation function is then used to obtain probability estimates for each emotion.
- The model goes through fine-tuning on the training set with the use of a **weighted loss function** to reduce class imbalance across all emotion categories. The performance of the fine-tuned model is tracked on the validation set through **averaged micro and macro F1 scores**, which are known to be standard metrics for multi-label classification.

## Decision Threshold Optimization

- After having trained the model, it outputs each label's probability scores. This is done by using per-label decision thresholds instead of a single global decision threshold, due to the dataset imbalance present in the dataset. Additionally, these thresholds are then optimized with the use of the validation set.
- Multiple threshold values are then evaluated, with the one that obtains the highest F1 score being chosen. This procedure applies to every label individually.
- Regarding the threshold optimization, it is used as a step executed after processing, which does not affect the model parameters. The optimized

thresholds are then used to determine the final predicted emotions set during inference.

## Semantic Filtering and Selection

- To support the justification, a semantic retrieval component is built with the use of texts from the training set. Each one is embedded into a high-dimensional vector space through a pretrained sentence embedding model. Measuring semantic similarity between the formed vectors (correspondingly representing the comments) can be done with **cosine similarity**.
- Given an input comment, the model computes its embedding and uses it to retrieve semantically similar comments from the training set. Retrieval only applies to the training set to avoid data leakage.

## Retrieval Filtering and Selection

- The filtering phase ensures that the retrieved examples are semantically relevant and emotionally consistent with the emotions which were predicted in the input comment.
- The retrieved posts are filtered depending on the related emotion labels. Any comment that has at least one predicted emotion in common with the input comment is retained for filtering. After obtaining the set of similar comments, the few top-ranked examples with the best semantic similarity results are selected.

## Retrieval-Augmented Justification Generation

- Justifications are generated with a RAG framework.
  - A generation model receives the input comment, as well as its predicted emotion labels and the respective selected retrieved examples, grouping the three pieces of information and taking them as an input.
  - The generator then produces an explanation in natural language which connects emotional signals from the input comment to patterns which were detected in the retrieved comments.
- The justification is generated after this procedure ends. This clarifies the separation between prediction and explanation.

## Inference Pipeline

- Finally, inference consists of using a trained model to predict or make decisions based on unseen data. At this time, a comment is processed through the entire pipeline.
  1. The procedure starts with the emotion classifier predicting a set of emotions with the use of the optimized decision thresholds.
  2. Next, the similar comments are retrieved and filtered based on the emotions predicted in the input comment.
  3. Finally, justification is generated thanks to the RAG framework.
- This pipeline optimizes transparency and interpretability, which is the result of the clear separation between prediction and justification (as mentioned in the previous subsection).

# Experimental Results

## Evaluation Setup

- The classification model is evaluated on the splits of the dataset separately linked to testing and validation.
- Performance is measured by using the average F1 scores, respectively micro-average and macro-average.
  - Micro-F1 score focuses on prediction performance across all the noted labels
  - Macro-F1 analyses the same performance but focused on the least common emotion labels.
- Furthermore, two additional settings are considered:
  - A **baseline configuration**, by setting the global decision threshold as 0.5 (maximum uncertainty point, positive if the F1 score is above 0.5) for every label;
  - An optimized configuration with the use of per-label decision thresholds which were selected for validation.

## Quantitative Results

The following table shows the classification performance without optimization and with optimization after having fine-tuned RoBERTa: (*Check Figure 1 in the "Figures" section)*

- As visible, per-label decision threshold optimization significantly improves F1 scores. However, baseline configuration reveals efficient performance of the transformer-based model for multi-label emotion detection following fine-tuning. Furthermore, the results indicate that one global threshold is not enough for handling the class imbalance present in the dataset and that threshold tuning is indeed an effective strategy after fine-tuning.

## Qualitative Results: Retrieval-Augmented Justification

The system also investigates the effectiveness of the retrieval-augmented justification mechanism used in the project.

- As mentioned in previous sections, the model takes an input post and predicts a set of emotions based on the input, as well as retrieving semantically similar posts from the training set which have emotions in common.

The figure below shows an example where the model predicted emotions such as sadness and disappointment. The example sentence is *"I feel empty and exhausted, nothing makes sense anymore."*: *(Check Figure 2 in the "Figures" section)*

- As visible in the figure, the retrieved posts contain at least one emotion which is also present in the emotion prediction of the input post.
- As for the justification, the system selects some of the retrieved posts and use them as reasoning, since they show the same energy in terms of emotions as the input post.

These results show that retrieval-based explanations improve interpretability by grounding predictions in existent social media posts uploaded by humans. The system does this through reasoning patterns, by making predictions based on analogy.

# Conclusion

The project focuses on emotion detection on social media.

- The objective is achieved by using a **transformer-based classification model (RoBERTa)** and a **retrieval-augmented justification framework**. With the GoEmotions dataset being the target, the RoBERTa model was fine-tuned for the task of multi-label emotion classification, which performed efficiently on the validation set and the test set. More specifically, the **optimization of the per-label decision threshold** brought impactful improvements in averaged F1 scores overall, revealing the significant effectiveness of this procedure in dealing with class imbalance.

- Not only does the system perform accurately when it comes to predictions, but it also **improves interpretability**, with the help of retrieval-augmented justification.

    - In this project, the system provides example-based justifications by retrieving social media posts with the most semantically similar meaning. In other words, the example posts which contain emotion labels which are also present in the input post are used as reasoning and justification. This procedure solidifies predictions made by models in the context of human-written text by providing explanations based on evidence which are interpretable in a simple way.

    - It is important to note that the justification mechanism applied in the project is used **after** having trained the transformer-based classification model, therefore not influencing the classification performance, which marks the separation between prediction and explanation.

# Future Work

There are many aspects which can be improved.

1. The quality of justification can be increased with the generation of short natural language explanations that align predicted emotions to retrieved evidence. With the already existing examples as evidence, this addition boosts **interpretability** and **understanding**.

2. Other retrieval approaches can obtain better results. Approaches such as emotion-aware or diversity-based retrieval can be explored to provide justifications with more information and

variety to the already existent sources of evidence. Furthermore, the evaluation of justification quality through human studies offers important insight into the relevance and readability of the displayed explanation from the point of view of a user.

3. Extending the technical part of the project to different domains such as mental health support systems can show the flexibility of retrieval-augmented justification beyond social media emotion detection. This extends the use of the justification framework to more complete and active fields.

4. Statistics obtained from the procedure of the project can be optimized if necessary. For example, the final macro-averaged and micro-averaged F1 scores obtained from fine-tuning RoBERTa and threshold tuning are greater than the initial model's documented average F1 score but can also be significantly higher.

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I was given clear instructions on the procedure of the project during its development. Communication was efficient and consistent. Availability was agreed between both and acted upon when required.

I was comfortable reaching out at any time when necessary and I am very satisfied with the environment I experienced throughout the entire project.

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## FIGURES

| Setting | DEV Micro-F1 | DEV Macro-F1 | TEST Micro-F1 | TEST Macro-F1 |
|---|---|---|---|---|
| Fixed threshold (0.5) | 0.510 | 0.469 | 0.501 | 0.453 |
| Tuned per-label thresholds | 0.598 | 0.540 | 0.597 | 0.521 |

Figure 1: classification performance without optimization and with optimization after having fine-tuned RoBERTa

```
Predicted emotions: ['annoyance', 'disappointment', 'nervousness', 'sadness', 'neutral']

Retrieved similar posts (evidence):
(Ex1) labels=['annoyance', 'disappointment', 'sadness'] | Virtually no happiness anymore, exhausted with trying, meds not working properly and just tired of it
(Ex2) labels=['disappointment'] | Living is exhausting....
(Ex3) labels=['neutral'] | I've isolated myself to the extent that i only focus my energy on myself/work.as lonely as it sounds its been such a bre...
(Ex4) labels=['neutral'] | It's like how I feel after I eat McDonalds. Satisfied, disappointed, sad, full, tired....
(Ex5) labels=['sadness'] | I'm kind of struggling tonight. Bored, lonely... man...

Justification:
 [Ex4] It's like how I feel after I eat McDonalds. Satisfied, disappointed, sad, full, tired. [Ex5] I'm kind of struggling tonight. Bored, lonely... man
```

Figure 2: Example of Retrieval-augmented justification